

ANALYTIC GEOMETRY

Polar Coordinates, Distance Formula, Area of Triangle & Conic Sections

Course Code: MATMJ52 — Platform: www.sciqb.com

1 Introduction to Polar Coordinates

In two-dimensional geometry, a point in a plane can be uniquely represented using a distance and an angle relative to a fixed reference point and direction. This system is known as the **Polar Coordinate System**.

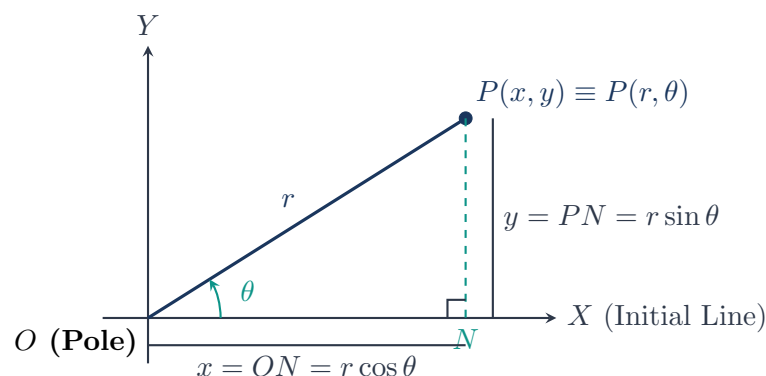
Definitions: Components of Polar Coordinates

- **Pole (or Origin) $[O]$:** A fixed reference point in the plane, denoted by O .
- **Initial Line $[OX]$:** A fixed horizontal straight line emanating from the pole O extending to the right.
- **Radius Vector $[r]$:** The line segment OP connecting the pole O to any point P in the plane. Its length is denoted by $r = |OP| \geq 0$.
- **Vectorial Angle $[\theta]$:** The angle measured counter-clockwise from the initial line OX to the radius vector OP . It is denoted by $\theta = \angle XOP$.

Thus, any point P is represented by the ordered pair (r, θ) .

1.1 Transformation between Cartesian and Polar Coordinates

Let $P(x, y)$ be a point in the Cartesian coordinate system with the origin coinciding with the pole O and the positive x -axis coinciding with the initial line OX . Drop a perpendicular PN from P onto OX .



From the right-angled triangle $\triangle ONP$:

1. Cartesian in terms of Polar Coordinates:

$$ON = x = r \cos \theta \quad \text{and} \quad PN = y = r \sin \theta \quad (1)$$

2. **Polar in terms of Cartesian Coordinates:** Squaring and adding equations (1):

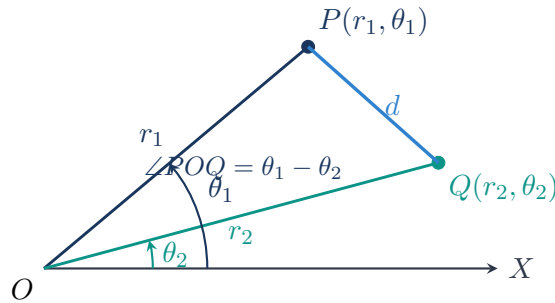
$$x^2 + y^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2 \implies r = \sqrt{x^2 + y^2} \quad (2)$$

Dividing y by x :

$$\tan \theta = \frac{PN}{ON} = \frac{y}{x} \implies \theta = \tan^{-1} \left(\frac{y}{x} \right) \quad (3)$$

2 Distance Between Two Points in Polar Coordinates

Let $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$ be two given points in polar coordinates with respect to the pole O and initial line OX .



In $\triangle POQ$:

- $OP = r_1, \quad OQ = r_2$
- $\angle POQ = |\theta_2 - \theta_1|$

By applying the **Law of Cosines** in $\triangle POQ$:

$$PQ^2 = OP^2 + OQ^2 - 2 \cdot OP \cdot OQ \cdot \cos(\angle POQ)$$

$$PQ^2 = r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)$$

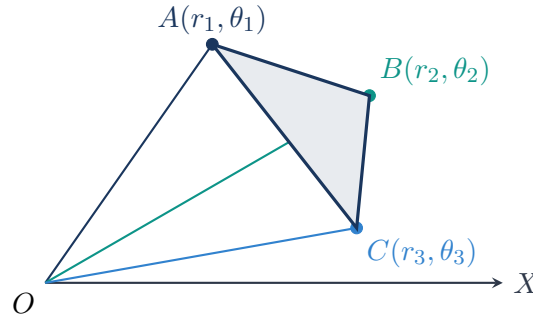
Polar Distance Formula

The distance $d = PQ$ between two points $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$ is given by:

$$PQ = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)} \quad (4)$$

3 Area of a Triangle in Polar Coordinates

Consider a triangle $\triangle ABC$ with vertices in polar coordinates given by $A(r_1, \theta_1)$, $B(r_2, \theta_2)$, and $C(r_3, \theta_3)$.



The area of $\triangle ABC$ can be obtained by summing/subtracting the areas of triangles formed with the pole O :

$$\text{Area}(\triangle ABC) = \text{Area}(\triangle AOB) + \text{Area}(\triangle AOC) - \text{Area}(\triangle BOC)$$

Recall that the area of any triangle with sides a, b and included angle α is $\frac{1}{2}ab \sin \alpha$. Therefore:

$$\text{Area}(\triangle AOB) = \frac{1}{2}r_1r_2 \sin(\theta_2 - \theta_1)$$

$$\text{Area}(\triangle BOC) = \frac{1}{2}r_2r_3 \sin(\theta_3 - \theta_2)$$

$$\text{Area}(\triangle AOC) = \frac{1}{2}r_1r_3 \sin(\theta_3 - \theta_1) = -\frac{1}{2}r_1r_3 \sin(\theta_1 - \theta_3)$$

Combining these yields the general area formula:

Area of Triangle in Polar Coordinates

The area of triangle $\triangle ABC$ with vertices $A(r_1, \theta_1), B(r_2, \theta_2), C(r_3, \theta_3)$ is:

$$\text{Area}(\triangle ABC) = \frac{1}{2} \left| r_1r_2 \sin(\theta_2 - \theta_1) + r_2r_3 \sin(\theta_3 - \theta_2) + r_3r_1 \sin(\theta_1 - \theta_3) \right| \quad (5)$$

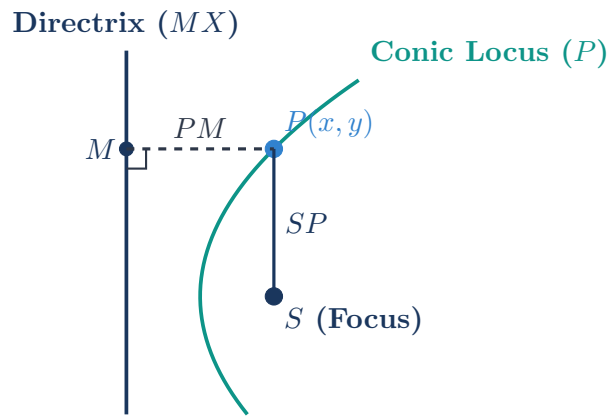
4 Polar Equation of a Conic Section

Definition: Conic Section

A **conic section** (or simply conic) is defined as the locus of a moving point P in a plane such that the ratio of its distance from a fixed point S (called the **focus**) to its perpendicular distance from a fixed straight line MX (called the **directrix**) is always constant.

This constant ratio is called the **eccentricity** of the conic, denoted by e :

$$\frac{SP}{PM} = e \quad (6)$$



4.1 Classification of Conics based on Eccentricity (e)

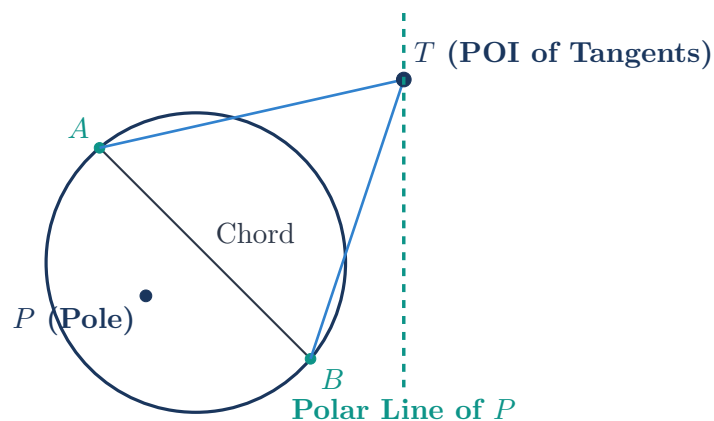
Depending on the value of eccentricity e , the locus of point P forms three fundamental conics:

1. **Parabola** ($e = 1$): Here $SP = PM$. The locus is a parabola.
2. **Ellipse** ($e < 1$): Here $SP < PM$. The locus is an ellipse.
3. **Hyperbola** ($e > 1$): Here $SP > PM$. The locus is a hyperbola.

5 Theory of Pole and Polar

Definition: Pole and Polar

If chords of a conic section are drawn through a fixed point P , and tangents are drawn at the extremities (A and B) of all such chords, then the locus of the point of intersection of these two tangents is a straight line called the **Polar** of the fixed point P . The fixed point P itself is called the **Pole** of that straight line.



6 Summary of Key Formulas

Analytic Geometry Quick Reference Sheet

1. **Cartesian to Polar:** $x = r \cos \theta$, $y = r \sin \theta$

2. **Polar to Cartesian:** $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1} \left(\frac{y}{x} \right)$

3. **Polar Distance Formula:** $d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)}$

4. **Area of Polar Triangle:**

$$\text{Area} = \frac{1}{2} \left| r_1 r_2 \sin(\theta_2 - \theta_1) + r_2 r_3 \sin(\theta_3 - \theta_2) + r_3 r_1 \sin(\theta_1 - \theta_3) \right|$$

5. **Conic Eccentricity Ratio:** $\frac{SP}{PM} = e$ (Parabola: $e = 1$, Ellipse: $e < 1$, Hyperbola: $e > 1$)